Assignment No - 4

Assignmen No - 1

① Print an nxn start square
Take a dynamic input from the user & print an nxn square.

ex -

```
* * * *
* * * *
* * * *
* * * *
```

```
PSVM ()
{
int n=4;
    for(int i=1; i<n; i++){
        for(int j=1; j<n; j++){
                Sop("*");
        }
    }
    Sopln();
}
}
```

TC - $O(n^2)$
SC - $O(1)$

---

② Print the following pattern

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

Code -

```
Class Demo {
        PSVM ()
        {
        int n=4
        for(int i=1; i<n; i++){
```

```
for(int j=1; j<n; j++){
            Sop(j);
        }
        Sopln();
    }
}
}
```

TC – $O(n^2)$
SC – $O(1)$

Q.3 print the following pattern

for n=4

```
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

Code –

```
Class Demo{
    Psvm(){
        int n=4;
        for(int i=1; i<n; i++){
            for(int j=1; j<n; j++){
                Sop(j);
            }
            Sopln();
        }
    }
}
```

TC – $O(n^2)$
SC – $O(1)$

Assignment-2 -

① Hollow Square -

print a hollow squre of size n

If n=5

```
* * * * *
*       *
*       *
*       *
* * * * *
```

Code -

```
Class Demo{
    PSVM (){
        int n=5;
    for(int i=1; i<n; i++){
        for(int j=1; j<n; j++){
            if(i==1||j==1||j==5||j==5){
                Sop("*" + " ");
            }else{
                sop(" ");
            }
        }
        Sopln();
    }
}
}
```

TC -	$O(n^2)$
SC -	$O(1)$

② Diagonal Square (X pattern)
print the following pattern
if n=5

```
*   *
 * *
  *
 * *
*   *
```

code -

```
class Demo{
    psvm(){
        int n=5;
    for(int i=1; i<n; i++){
        for(int j=1; j<n; j++){
            if(i==j || i+j==6){
                Sop("*");
            }else{
                Sop(" ");
            }
            }
        Sopln();
    }
    }
}
```

TC —	$O(n^2)$
SC —	$O(1)$

③ Right Triangle

If n=5

```
*
* *
* * *
* * * *
* * * * *
```

Code

```
class Demo{
    psvm(){
        int n=5;
    for(int i=1; i<=n; i++){
        for (int j=1; j<=i; j++){
            Sop("*" + " ");
        }
        Soptln();
    }
}
}
```

TC –	$O(n^2)$
SC –	$O(1)$

---

④ Inverted Right Triangle –

If n=5

```
* * * * *
* * * *
* * *
* *
*
```

Code –

```
class Demo{
    psvm(){
        int n=5;
    for(int i=n; i>=1; i--){
```

```
for(int j=1; j<i ; j++){
        sopln("*" + " ");
    }
    Sopln();
  }
}
```

TC –	$O(n^2)$
SC –	$O(1)$

---

(5) Chessboard pattern

If n=4

```
* # * #
# * # *
* # * #
# * # *
```

Code –

```
Class Demo{
    PSVm(){
        int n=5;
    for(int i=1; i<n; i++){
        for(int j=1; j<n; j++){
            if((i+j)%2==0){
                sop("*" + " ");
            }else{
                sop("#" + " ");
            }
        }
        Sopln();
    }
}
```

TC –	$O(n^2)$
SC –	$O(1)$

Assignment No-3

① Double Number pyramid

for n=4

```
1
1 2
1 2 3
1 2 3 4
1 2 3
1 2
1
```

code -

```
class Demo{
    psvm(){
        int n=4
upper case.-
        for(int i=1; i<n; i++){
            for(int j=1; j<n-i; j++){
        }
        for(int j=1; j<i; j++){
                Sop(j+" ");
        }
        Sopln;
    }
```

TC -	$O(n^2)$
SC -	$O(1)$

Lower case-

```
        int n=3;
    for(int s=n; s>=1; s--){
        for(int p=1; p<n; p++){
            Sop(" ");
        }
        for(int p=1; p<s; p++){
```

```
Sop ("" + p);
}
Sopln();
}
}
}
```

TC –	$O(n^2)$
SC –	$O(1)$